STA2002 Tutorial 2

1. Review

Suppose we have a sequence of iid random variables $\{X_1, ..., X_n\}$ with pmf/pdf $f(x;\theta)$. **How to estimate** θ?

The **maximum likelihood estimator (MLE)** $\hat{\theta}$ is an maximizer of the likelihood

$$\hat{\theta} = \arg\max_{\theta\in\Theta} L(\theta; x_1, ..., x_n) \equiv \arg\max_{\theta\in\Theta} \prod_{i=1}^{n} f(x_i;\theta)$$

$$= \arg\max_{\theta\in\Theta} l(\theta; x_1, ..., x_n) \equiv \arg\max_{\theta\in\Theta} \log L(\theta; x_1, ..., x_n) = \arg\max_{\theta\in\Theta} \sum_{i=1}^{n} \log f(x_i;\theta)$$

The sufficient condition that $\hat{\theta}$ is an MLE estimator is that

$$\left.\frac{\partial l(\theta; x_1, ..., x_n)}{\partial\theta}\right|_{\theta=\hat{\theta}} = 0 \quad \text{and} \quad \left.\frac{\partial^2 l(\theta; x_1, ..., x_n)}{\partial^2\theta}\right|_{\theta=\hat{\theta}} < 0.$$

In the case that the sufficient condition for MLE estimator is hard to solve, we can find another estimator via **method of moments**, i.e., for parameter $\theta = (\theta_1, ..., \theta_d)$:

- Derive the theoretical moments as functions of θ

$$E[X^k] = g_k(\theta_1, ..., \theta_d), \quad \text{for } k = 1, ..., d.$$

- Estimate the sample moments from data as

$$M_k = \frac{1}{n}\sum_{i=1}^{n} X_i^k.$$

- The **method of moments estimator** $\hat{\theta}$ is the solution of the equations

$$g_k(\theta_1, ..., \theta_d) = M_k, \quad \text{for } k = 1, ..., d.$$

Suppose we have an estimator $\hat{\theta}$ for some parameter θ. **How to evaluate** $\hat{\theta}$? We call an estimator is **unbiased** if

$$\mathbb{E}[\hat{\theta}] = \theta.$$

The **mean squared error (MSE)** of $\hat{\theta}$ is defined as

$$MSE = \mathbb{E}[\hat{\theta} - \theta]^2 = Var[\hat{\theta}] + \Big(\mathbb{E}[\hat{\theta}] - \theta\Big)^2.$$

PS: Some `R` programming is needed for this course. We would provide some necessary document on blackboard. Please come to our office hour if you could not install `R` successfully.

1. Let $X_1, X_2, \cdots, X_n$ be an independent random sample from the p.d.f. given by

$$f(x;\theta) = \frac{1}{\theta} x^{(1-\theta)/\theta}, \quad 0 < x < 1, 0 < \theta < \infty$$

(a) Find the MLE of θ.

(b) Show that the MLE is an unbiased estimator of θ.

(a)

$$L(\theta) = \left(\frac{1}{\theta^n}\right)\left(\prod_{i=1}^{n} x_i\right)^{1/\theta-1}, \quad 0 < \theta < \infty,$$

$$\ln L(\theta) = -n \ln \theta + \left(\frac{1}{\theta} - 1\right) \ln \prod_{i=1}^{n} x_i,$$

$$\frac{d \ln L(\theta)}{d\theta} = \frac{-n}{\theta} - \frac{1}{\theta^2} \ln \prod_{i=1}^{n} x_i = 0,$$

$$\hat{\theta} = -\frac{1}{n} \ln \prod_{i=1}^{n} x_i = -\frac{1}{n} \sum_{i=1}^{n} \ln x_i,$$

$$\left.\frac{d^2 \ln L(\theta)}{d\theta^2}\right|_{\theta=\hat{\theta}} = -\frac{n}{\hat{\theta}^2} < 0.$$

(b) We first find $E(\ln X)$

$$E(\ln X) = \int_0^1 \ln x (1/\theta) x^{1/\theta - 1} dx.$$

Using integration by parts,

$$E(\ln X) = \lim_{a \to 0} \left[x^{1/\theta} \ln x - \theta x^{1/\theta}\right]_a^1 = -\theta.$$

Thus,

$$E(\hat{\theta}) = -\frac{1}{n} \sum_{i=1}^{n} (-\theta) = \theta.$$

2. Suppose X has a Gamma distribution with parameters α and β. Find the method-of-moments estimator of the parameters.

   Because we have two parameters, the method of moments methodology requires us to determine the first two moments.

$$\mathrm{E}\left(X^k\right) = \int_{x=0}^{\infty} x^k \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x} dx = \frac{\Gamma(\alpha+k)}{\Gamma(\alpha)} \frac{1}{\beta^k} \int_{x=0}^{\infty} \frac{\beta^{\alpha+k}}{\Gamma(\alpha+k)} x^{\alpha+k-1} e^{-\beta x} dx = \frac{\Gamma(\alpha+k)}{\Gamma(\alpha)} \frac{1}{\beta^k}.$$

   Therefore,

$$\mu_1 = \mathrm{E}(X) = \frac{\alpha}{\beta}, \quad \mu_2 = \mathrm{E}\left(X^2\right) = \frac{\alpha(\alpha+1)}{\beta^2} = \frac{\alpha^2}{\beta^2} + \frac{\alpha}{\beta^2}.$$

   Using method of moments, we set

$$\frac{\alpha}{\beta} = \overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i,$$

$$\frac{\alpha^2}{\beta^2} + \frac{\alpha}{\beta^2} = \overline{X^2} = \frac{1}{n}\sum_{i=1}^{n} X_i^2.$$

   Therefore, we obtain the estimators as:

$$\hat{\beta} = \frac{\bar{X}}{\overline{X^2} - (\bar{X})^2} \quad \text{and} \quad \hat{\alpha} = \hat{\beta}\bar{X} = \frac{(\bar{X})^2}{\overline{X^2} - (\bar{X})^2}.$$

3. Let $X_1, X_2, \ldots, X_n$ be a random sample of size n from a uniform distribution on the interval $[\theta - 1, \theta + 1]$
   (a) Find the method of moments estimator of θ. ls this estimator an unbiased estimator of θ ?
   (b) Find the MLEs of θ.

(a) The mean of X is

$$E(X) = \frac{(\theta - 1) + (\theta + 1)}{2} = \theta.$$

Thus, the method of moments estimator of θ is $\hat{\theta}_{Mom} = \bar{X}$.

$$E\left(\hat{\theta}_{Mom}\right) = E(\bar{X}) = \theta,$$

i.e., this estimator is unbiased.
(b) The likelihood function is

$$L\left(\theta; x_1, \ldots, x_n\right) = \begin{cases} (1/2)^n & \text{if all } x_i \in [\theta - 1, \theta + 1], \\ 0 & \text{otherwise.} \end{cases}$$

To maximize L, θ should be in the range of

$$\theta - 1 \leq X_{(1)} \leq X_{(2)} \leq \ldots \leq X_{(n)} \leq \theta + 1,$$

then, $L = (1/2)^n$. Thus, any $\hat{\theta}_{mle} \in \left[X_{(n)} - 1, X_{(1)} + 1\right]$ is a MLE for θ.

4. Suppose $X_1, X_2, \ldots, X_n$ are i.i.d. normal random variables, with mean μ and variance σ^2. Both μ and σ^2 are unknown. Here are two estimators for σ^2:

$$\tilde{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2,$$
$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2 = \frac{n-1}{n}\tilde{\sigma}^2.$$

In the lecture, we showed that $\hat{\sigma}^2$ is the MLE of σ^2. We evaluate these two estimators in the following questions.

(a) Derive the mean of $\hat{\sigma}^2$ and $\tilde{\sigma}^2$. Hint: $\sum_{i=1}^{n}(X_i - \bar{X})^2/\sigma^2 = \boldsymbol{x}^T(I - \frac{1}{n}\mathbf{1}\mathbf{1}^T)\boldsymbol{x}/\sigma^2 \sim \chi^2(n-1)$.

(b) Derive the variance of $\hat{\sigma}^2$ and $\tilde{\sigma}^2$.

(c) Compare the mean squared error (MSE) of $\hat{\sigma}^2$ and $\tilde{\sigma}^2$. To minimize MSE, which one do we prefer?

Solution: Recall that $\sum_{i=1}^{n}(X_i - \bar{X})^2/\sigma^2 \sim \chi^2(n-1)$, with mean $n-1$ and variance $2(n-1)$.

(a)

$$E(\tilde{\sigma}^2) = E\left(\frac{\sigma^2}{n-1}\frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{\sigma^2}\right) = \sigma^2,$$
$$E(\hat{\sigma}^2) = E\left(\frac{n-1}{n}\tilde{\sigma}^2\right) = \frac{n-1}{n}\sigma^2.$$

(b)

$$Var(\tilde{\sigma}^2) = Var\left(\frac{\sigma^2}{n-1}\frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{\sigma^2}\right) = \frac{\sigma^4}{(n-1)^2}2(n-1) = \frac{2\sigma^4}{n-1},$$
$$Var(\hat{\sigma}^2) = Var\left(\frac{n-1}{n}\tilde{\sigma}^2\right) = \frac{(n-1)^2}{n^2}\frac{2\sigma^4}{n-1} = \frac{2(n-1)\sigma^4}{n^2}.$$

(c) The difference between MSE is:

$$\begin{aligned}
\mathrm{MSE}\left(\tilde{\sigma}^2\right) - \mathrm{MSE}\left(\hat{\sigma}^2\right) &= \left(\mathrm{E}\left(\tilde{\sigma}^2\right) - \sigma^2\right)^2 + \mathrm{Var}\left(\tilde{\sigma}^2\right) - \left[\left(\mathrm{E}\left(\hat{\sigma}^2\right) - \sigma^2\right)^2 + \mathrm{Var}\left(\hat{\sigma}^2\right)\right] \\
&= 0 + \mathrm{Var}\left(\tilde{\sigma}^2\right) - \left[\left(\frac{n-1}{n}\sigma^2 - \sigma^2\right)^2 + \left(\frac{n-1}{n}\right)^2 \mathrm{Var}\left(\tilde{\sigma}^2\right)\right] \\
&= -\sigma^4\left(\frac{n-1}{n} - 1\right)^2 + \mathrm{Var}\left(\tilde{\sigma}^2\right)\left[1 - \left(\frac{n-1}{n}\right)^2\right] \\
&= \sigma^4 \cdot \frac{3n-1}{(n-1)n^2} > 0.
\end{aligned}$$

Therefore, we will prefer $\hat{\sigma}^2$ as our estimator.